

Retatrutide

Extensively Studied

Triple GLP-1/GIP/Glucagon Agonist | Weight Loss & Diabetes

Injectable

Typical Dose

0.5-12 mg weekly

Once weekly

Route

Injectable

Abdomen, thigh, upper arm - rotate weekly to prevent...

Cycle

Continuous therapy

Typical duration

Storage

2-8°C

Refrigerated

Did you know? You can suggest edits to improve this peptide information. [Log in to contribute](#)

Overview

What is Retatrutide?

Retatrutide (LY3437943) is a novel triple hormone receptor agonist targeting GLP-1, GIP, and glucagon receptors. Phase III TRIUMPH-4 trial (Dec 2025) achieved 28.7% weight loss (71.2 lbs) at 68 weeks with 12mg dose - the highest recorded for any obesity medication. Currently in late-stage Phase III development by Eli Lilly with FDA approval expected 2026-2027.

Key Benefits

Triple hormone receptor activation provides superior weight loss (24.2%), improved glycemic control, and enhanced cardiovascular benefits compared to single or dual agonists

Mechanism of Action

Activates GLP-1 for appetite suppression, GIP for insulin sensitivity, and glucagon for increased energy expenditure and hepatic fat oxidation

Molecular Information

Weight	Length	Type
4,731.33 Da	39 amino acids	Triple GLP-1/GIP/glucagon agonist

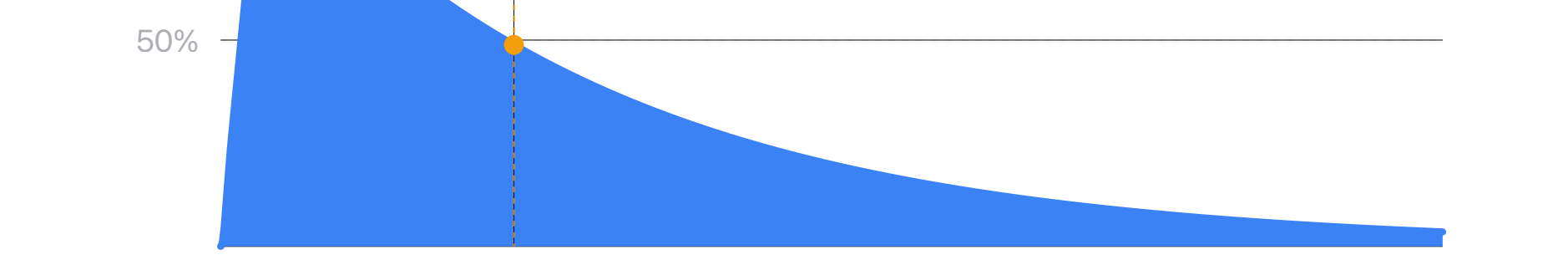
Amino Acid Sequence:

His-Ala-Glu-Gly-Thr-Phe-Thr-Ser-Asp-Val-Ser-Ser-Tyr-Leu-Glu-Gly-Gln-Ala-Ala-Lys-Glu-Phe-Ile-Ala-Trp-Leu-Val-Arg-Gly-Arg-Gly-Pro-Ser-Ser-Ser-Gly-Ala-Pro-Pro-Ser

* C20 fatty acid conjugation with Alb residues for DPP-4 resistance

Pharmacokinetics

Peak: 1 day Half-life: 6 days Cleared: ~30 days



Research Indications

Weight Loss

Most Effective

Superior Weight Reduction

Clinical trials demonstrate 17.5% at 24 weeks and 24.2% at 48 weeks - highest recorded for any obesity medication in development

Sustained Weight Management

Continuous weight loss throughout trials with no plateau reached at 48 weeks, suggesting greater long-term potential than current therapies

Triple Mechanism Obesity Treatment

Addresses obesity through appetite suppression, increased energy expenditure, and improved metabolic efficiency via three hormone pathways

Diabetes

Effective

Cardiovascular

Moderate

Research Protocols

Disclaimer: These are commonly discussed research protocols and not medical advice. Consult a healthcare provider before use.

Goal	Dose	Frequency	Route
Conservative Starting Dose (Week 1-4)	0.5 mg weekly	Once weekly	Subcutaneous injection
Low Maintenance Dose (Week 4-8)	1 mg weekly	Once weekly	Subcutaneous injection
Standard Escalation (Week 8-12)	2 mg weekly	Once weekly	Subcutaneous injection
Moderate Weight Loss (Week 12-16)	4 mg weekly	Once weekly	Subcutaneous injection
Advanced Weight Loss (Week 16-20)	8 mg weekly	Once weekly	Subcutaneous injection
Maximum Efficacy (Week 20+)	12 mg weekly	Once weekly	Subcutaneous injection
Type 2 Diabetes - Conservative Start	0.5-1 mg weekly	Once weekly	Subcutaneous injection
Type 2 Diabetes - Maintenance	4-8 mg weekly	Once weekly	Subcutaneous injection
Clinical Trial Protocol (Obesity Study)	1-2 mg weekly start	Once weekly	Subcutaneous injection

Timing: Administer on the same day each week for consistent hormone regulation. Can be taken with or without food due to subcutaneous administration.

Peptide Interactions

Tirzepatide	Avoid Combination
Cagrilintide	Use Caution
Semaglutide	Avoid Combination
Insulin	Monitor Combination
Metformin	Compatible
SGLT2 Inhibitors	Compatible
Oral Contraceptives	Requires Timing
Warfarin	Monitor Combination
BPC-157	Compatible
NAD+	Compatible

How to Reconstitute

Calculator

Important: Use BAC water only. Avoid BAC saline (sodium chloride) - may cause precipitation or cloudiness. Refrigerate and use within 28 days.

- Remove Retatrutide vial from refrigeration and allow to reach room temperature (15-20 minutes)
- Clean vial top with alcohol swab and allow to air dry completely
- Add calculated amount of bacteriostatic water slowly down the side of the vial to minimize foaming
- Gently swirl the vial in circular motions - DO NOT shake vigorously as this may damage the peptide structure
- Allow to fully dissolve (may take 2-3 minutes) - solution should be clear and colorless
- Store reconstituted solution in refrigerator (2-8°C) and use within 28 days
- Draw calculated dose using insulin syringe, inject subcutaneously into abdomen, thigh, or upper arm
- Rotate injection sites weekly to prevent lipodystrophy and maintain absorption consistency

Quality Indicators

- Pharmaceutical-grade white powder**
Lyophilized Retatrutide should appear as a white to off-white powder with uniform texture and no discoloration
- Proper cold chain maintenance**
Reconstituted solution requires consistent refrigeration at 2-8°C with temperature monitoring
- Clear reconstituted solution**
Properly reconstituted Retatrutide forms a clear, colorless solution without particles, cloudiness, or precipitates
- Stable extended half-life effects**
Consistent appetite suppression and metabolic effects between weekly doses indicate proper potency
- Source verification critical**
Due to investigational status and high demand, counterfeit versions circulate - verify pharmaceutical-grade sourcing
- Rapid tolerance or effectiveness loss**
Genuine Retatrutide maintains efficacy long-term; rapid tolerance suggests degraded or counterfeit product
- Unusual side effect profile**
Effects significantly different from expected GI-predominant profile may indicate contaminated or incorrect product

What to Expect

- Week 1-2: Initial appetite suppression and mild GI effects as body adapts to triple hormone activation
- Week 2-4: Noticeable reduction in food cravings and portion sizes, early weight loss (2-5%)
- Week 4-8: Significant appetite control and steady weight loss (5-10%), improved glucose control if diabetic
- Week 8-16: Substantial weight reduction (10-18%) with enhanced energy expenditure and metabolic improvements
- Week 16-24: Major weight loss milestone (15-22%) with cardiovascular benefits and liver fat reduction
- Week 24-48: Maximum clinical efficacy (20-24.2%) with comprehensive metabolic improvements and sustained benefits

Side Effects & Safety

- Side Effects** 8 **When to Stop** 6
- Most common side effects are gastrointestinal (nausea 43%, diarrhea 33% at 12mg) - typically mild to moderate and dose-dependent
 - NEW SIGNAL (Dec 2025): Dysesthesia (abnormal touch sensations) reported in 8.8-20.9% of participants at 9-12mg doses in TRIUMPH-4 trial
 - Conservative start at 0.5mg weekly minimizes GI side effects (13% vs 73-94% at higher doses) - escalate gradually every 4 weeks
 - Phase III discontinuation rates: 12.2% (9mg) and 18.2% (12mg) due to adverse events - correlated with baseline BMI
 - Monitor for signs of pancreatitis (severe abdominal pain radiating to back) - discontinue immediately if suspected
 - Heart rate increases are common, especially in first 24 weeks - monitor cardiovascular status regularly
 - Contraindicated in patients with personal/family history of medullary thyroid carcinoma or MEN2 syndrome
 - May cause rapid weight loss - some discontinuations due to perceived excessive weight loss

References

- Research Studies** 7 **Citations** 9
- Phase III TRIUMPH-4 Trial - Eli Lilly (Dec 2025)**
Human | 9-12mg weekly | 68 weeks | 28.7% weight loss (71.2 lbs)
First successful Phase III trial in 445 adults with obesity and knee osteoarthritis. 12mg dose achieved 28.7% weight loss (71.2 lbs from 248.5 lb baseline) with 75.8% pain reduction. Discontinuation rates 12-18% due to GI effects; new dysesthesia signal in 8.8-20.9% of participants.
[View Study](#) →
- MASLD Substudy - Nature Medicine (2024)**
Human | 8-12mg weekly | 24 weeks | 82% liver fat reduction
Metabolic dysfunction-associated steatotic liver disease study showing dramatic liver fat reductions with normalization in over 90% of participants at highest doses.
[View Study](#) →
- TRIUMPH-Outcomes Cardiovascular Study (2024-2029)**
Human | 8mg weekly | 248 weeks | 10,000 participants with ASCVD
Major cardiovascular outcomes trial assessing effects on MACE and kidney outcomes in participants with obesity and established cardiovascular or kidney disease.
[View Study](#) →

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Quick Start Guide

- Typical Dose**
Start 0.5mg weekly (clinical practice) or 1mg weekly (trial protocol), escalate every 4 weeks to target 8-12mg
- How Often**
Once weekly
- Where to Inject**
Abdomen, thigh, upper arm - rotate weekly to prevent lipodystrophy
- Injection Timing**
Any time of day, maintain same day each week for consistency
- Effects Timeline**
24-48 hours: appetite suppression;
4-8 weeks: significant weight loss;
24+ weeks: maximum benefits
- Storage**
Lyophilized powder at room temperature; reconstituted solution refrigerated
- Cycle Length**
Continuous therapy - effects reverse upon discontinuation
- Break Between**
No cycling required - designed for continuous use

Help Us Gain Real Insights

Question 1 of 10

What is your experience with this compound?

☐ Currently using

☐ Used in the past

☐ Planning to start

☐ Just researching

☐ Other (please specify)

[Submit Answer](#)

Community Insights

5,592 responses

- Effectiveness**
92% report positive results
- Would Recommend**
88% of 455 users
- Common Side Effects**
Nausea, Fatigue, Headache
- Best Results At**
6-12 months (82% effective)

From 3123 users tracking

Weight Change

-2.8% avg

85% saw decrease

Sleep Duration

0h avg

19% saw increase

Resting Heart Rate

34% decrease, 51% increase

Was this helpful?

☒ Yes ☐ No

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